Subgroup

→ Let $(G, *)$ be a group and H be a non-empty subset of G. If $(H, *)$ is itself a group, then $(H, *)$ is called a subgroup of $(G, *)$.

→ If H be a subset of a group G such that
(i) the identity e of G belongs to H.
(ii) if a and b belong to H then $a * b \in H$
(iii) if $a \in H$, then $a^{-1} \in H$.
where H is called a subgroup of G.

E.g. Let $G = \{1, -1, i, -i\}$ and $H = \{1, -1\}$.
Here, G and H are groups with respect to the binary operation multiplication $\cdot$.
H is a subset of G and so, $(H, \cdot)$ is a subgroup of $(G, \cdot)$.

~~Ex.~~ 1 Let $(Z, +)$ be a group. Investigate on the following subsets of G
(i) the set G_1 of all odd integers.
(ii) the set G_2 of all positive integers
as subgroup of G.

Ans:-
(i) The set G_1 does not satisfy the closure law (property), since addition of two odd integers is always even. Thus, G_1 is not a subgroup of G.

(ii) Closure property :- The set G_2 is closed under the operation $+$, since addition of two even integers is even.

<u>Associative property</u> :- The operation + is associative on set of integers
$\therefore a+(b+c) = (a+b)+c \quad \forall\, a, b, c \in G_2$

<u>Identity property</u> :- The element $0 \notin G_2$
$\therefore (G_2, +)$ is not a subgroup of $(Z, +)$.

<u>IMP NOTE</u> :-

(i) If G is a group, then G is itself a subset of G. $\therefore (G, *)$ is also a subgroup of $(G, *)$.

(ii) Let $H = \{e\}$, where e is an identity element of Group $(G, *)$. Then $(H, *) = (\{e\}, *)$ is subgroup of G.

Thus, $(G, *)$ and $(\{e\}, *)$ are trivial subgroup of a Group $(G, *)$.

<u>Theorems</u> :-

① A non-empty subset H of a group G is a subgroup of G if and only if
(i) $a \in H, b \in H \Rightarrow a * b \in H$
(ii) $a \in H \Rightarrow a^{-1} \in H$ where a^{-1} is the inverse of an element a in G.

② Let $(G, *)$ be a group and $H (\neq \phi) \subseteq G$. Then $(H, *)$ is a subgroup of $(G, *)$ if and only if $a, b \in H \Rightarrow a * b^{-1} \in H$.

3. If $(H_1, *)$ and $(H_2, *)$ are both subgroups of the group $(G, *)$, then $(H_1 \cap H_2, *)$ is also a subgroup.

e.g. Let $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \ldots\}$ and let $H_1 = \{0, \pm 2, \pm 4, \pm 6, \ldots\}$ and $H_2 = \{0, \pm 3, \pm 6, \pm 9, \ldots\}$. Then $H_1 \cap H_2 = \{0, \pm 6, \pm 12, \ldots\}$. Here, $(H_1, +)$, $(H_2 +)$ are subgroups of $(\mathbb{Z}, +)$. $(H_1 \cap H_2, +)$ is also a subgroup of $(\mathbb{Z}, +)$.

But, $(H_1 \cup H_2, +)$ is not a subgroup of $(\mathbb{Z}, +)$. Because $H_1 \cup H_2 = \{0, \pm 2, \pm 3, \pm 4 \ldots\}$ and $2, 3 \in H_1 \cup H_2$ but $2+3 = 5 \notin H_1 \cup H_2$. Thus, $H_1 \cup H_2$ is not closed under $+$.

Example: Let H be a subset of a group G. Show that H is a subgroup of G if H has the following three properties.

(i) the identity element e belongs to H.

(ii) H is closed under the operation of G. i.e if $a, b \in H$ then $a * b \in H$.

(iii) H is closed under inverses. i.e. if $a \in H$ then $a^{-1} \in H$.

Cyclic Group.

→ A group G is said to be cyclic if it contains an element a such that every element of G is a power of a. The element a is said to generate G and we write $G = \langle a \rangle$.

If a generates G and all powers of a are distinct then

$$G = \{ \ldots, a^{-3}, a^{-2}, a^{-1}, e, a, a^2, a^3, \ldots \}$$

Here, G is an infinite cyclic group.

e.g: Group $(Z, +)$ is an infinite cyclic group.

$1^0 = 1$, $1^1 = 1$, $1^2 = 1+1 = 2$, $1^3 = 1+1+1 = 3$, ...

similarly, $1^{-1} = -1$, $1^{-2} = (1^2)^{-1} = -2$, ...

$Z = \langle 1 \rangle$

e.g. The multiplicative group $G = \{1, -1, i, -i\}$ is cyclic.

$G = \{ i, i^2, i^3, i^4 \}$ i.e. $G = \langle i \rangle$

$G = \{ -i, (-i)^2, (-i)^3, (-i)^4 \}$ i.e. $G = \langle -i \rangle$

$i^1 = i$

$i^2 = -1$

$i^3 = i^2 \cdot i$
$= -i$

$i^4 = i^2 \cdot i^2$
$= 1$

$(-i)^1 = -i$

$(-i)^2 = (-i)(-i) = +i^2 = -1$

$(-i)^3 = (-i)^2 \cdot (-i) = (-1)(-i)$
$= i$

$(-i)^4 = (-i^2)^2 = (-i)^2(-i)^2 = (-1)(-1)$
$= 1.$

Theorems of cyclic Groups

(1) Every cyclic group is an abelian.

(2) If a is a generator of a cyclic group G, then a^{-1} is also a generator of G.

(3) If a cyclic group G is generated by an element a of the order n, then a^m is a generator of G if and only if the greatest common divisor (g.c.d.) of m and n is 1. i.e. if and only if m and n are relative primes.

e.g. The set of integers $(Z, +)$ is a cyclic group of which the generator is 1. As $1^{-1} = -1$. -1 is also generator of $(Z, +)$. $Z = \langle 1 \rangle$, $Z = \langle -1 \rangle$.

e.g. The multiplicative group on $G = \{1, \omega, \omega^2\}$ is a cyclic group.

$\omega^0 = 1$, $\omega^1 = \omega$, $\omega^2 = \omega^2$, $\omega^3 = 1$, $\therefore G = \langle \omega \rangle$

$(\omega^2)^1 = \omega^2$, $(\omega^2)^2 = \omega^3 \cdot \omega = \omega$, $(\omega^2)^0 = 1$ $\therefore G = \langle \omega^2 \rangle$

ex How many generators are there in the cyclic group G of order 8?

Sol Let a be generator of G then $O(a) = 8$.

$\therefore G = \{a, a^2, a^3, \ldots, a^8\}$

Here, 7 is prime to 8 $\therefore a^7$ is ~~generator~~ generator of G.

5 is prime to 8 $\therefore a^5$ is a generator of G.

3 is prime to 8 $\therefore a^3$ is a generator of G.

$\therefore a, a^3, a^5, a^7$ are generators of G.

$\therefore$ There are 4 generators of G.

COSETS

→ Let $(G, *)$ be a group and H be any subgroup of G. Also, let a be any element of G, then the set

$$a * H = \{a * h : h \in H\}$$

is called the left coset of H in G generated by a. Similarly, the set

$$H * a = \{h * a : h \in H\}$$

is called the right coset of H in G generated by a.

If e is the identity element of G, then $H * e = H = e * H$. So, H itself is a right as well as a left coset.

If the group G is abelian, then we have $a * h = h * a$, $\forall h \in H$. Therefore, the right coset Ha will be equal to the corresponding left coset aH.

Properties of Cosets

Let H be a subgroup of G and $a, b \in G$, then,

(i) $a \in aH$

(ii) $aH = H$ if and only if $a \in H$

(iii) $aH = bH$ or $aH \cap bH = \phi$

(iv) $aH = bH$ if and only if $a^{-1} * b \in H$

Similar results holds for right cosets.

Ex $Z_{12} = \{0, 1, 2, \ldots 11\}$ $(Z_{12}, +)$
$H = \{0, 3, 6, 9\}$, $(H, +)$ is a subgroup of $(Z_{12}, +)$

$0+H = 0 + \{0, 3, 6, 9\} = \{0, 3, 6, 9\} = H$
$1+H = 1 + \{0, 3, 6, 9\} = \{1, 4, 7, 10\}$
$2+H = 2 + \{0, 3, 6, 9\} = \{2, 5, 8, 11\}$
Here, $0+H$, $1+H$ and $2+H$ are left cosets of H.

$H+0 = \{0, 3, 6, 9\} + 0 = \{0, 3, 6, 9\} = H$
$H+1 = \{0, 3, 6, 9\} + 1 = \{1, 4, 7, 10\}$
$H+2 = \{0, 3, 6, 9\} + 2 = \{2, 5, 8, 11\}$
Here, $H+0$, $H+1$ and $H+2$ are right cosets of H.
As left and right cosets are coincide, this subgroup H of Z_{12} is normal subgroup.

Ex Let $G = \{1, i, -1, -i\}$ & $H = \{1, -1\}$.
$(H, \cdot)$ is a subgroup of $(G, \cdot)$.

$i \cdot H = \{i \cdot 1, i \cdot (-1)\} = \{i, -i\}$
$1 \cdot H = \{1 \cdot 1, 1 \cdot (-1)\} = \{1, -1\}$

$H \cdot i = \{1 \cdot i, (-1) \cdot i\} = \{i, -i\}$
$H \cdot 1 = \{1 \cdot 1, (-1) \cdot 1\} = \{1, -1\}$

$i \cdot H$ and $1 \cdot H$ are left cosets and
$H \cdot i$ and $H \cdot 1$ are right cosets of H.
$(H, *)$ is a normal subgroup of $(G, \cdot)$.

Ex The symmetric group S_4 & $H = \{e, (12), (34), (12)(34)\}$ is an example where left and right subgroups are not coincide.
Hence, H is not normal subgroup of G.

Normal Subgroup

A subgroup H of a group G is called normal subgroup if $a^{-1}Ha \subseteq H$ for every $a \in G$. Equally, H is normal if $aH = Ha$ for every $a \in G$. i.e. the right and left cosets coincide.

Theorems of Normal subgroup

① Every subgroup of an abelian group is a normal subgroup.

② A subgroup H of a group G is normal if and only if $g^{-1}hg \in H$ for every $h \in H$, $g \in G$.

③ Let H be a normal subgroup of a group G, then the cosets of H form a group under coset multiplication

$$(aH)(bH) = abH$$

This group is called quotient group (or factor group) and is denoted by G/H. and it is denoted by G/H.

→ G/H has an identity element $H = eH$, for

$$(aH)(eH) = (ae)H = aH \quad \text{and}$$
$$(eH)(aH) = (ea)H = aH$$

Also, $a^{-1}H$ is the inverse of aH since

$$(a^{-1}H)(aH) = (a^{-1}a)H = eH = H \quad \text{and}$$
$$(aH)(a^{-1}H) = (aa^{-1})H = eH = H$$

→ If H is a subgroup of a group G, the number of distinct left or right cosets of H in G is called the index of H in G and is denoted by $[G:H]$ or by $i_G(H)$.

→ If G is a finite group and H is a normal subgroup of G, then

$$O(G/H) = O(G)/O(H) = i_G(H)$$

e.g. If $H = \{\ldots -6, -3, 0, 3, 6, \ldots\}$, then H_3 is a normal subgroup of $(Z, +)$.

The cosets of H in Z are $H_3 = \{\ldots -6, -3, 0, 3, 6, \ldots\}$, $1 + H_3 = \{\ldots -5, -2, 1, 4, 7, 10, \ldots\}$ and $2 + H_3 = \{\ldots -4, -1, 2, 5, 8 \ldots\}$. Therefore,

$G/H = \{H_3, 1+H_3, 2+H_3\}$ and $i_G(H) = 3$.

↳ Quotient group.

Ex Let $H = \{\ldots, -10, -5, 0, 5, 10, \ldots\}$ and $(H, +)$ is a subgroup of $(Z, +)$. Find (i) the cosets of H in Z and (ii) the index of H in Z.

Ans: There are five distinct (left) cosets of H in Z.

$0 + H = \{\ldots -10, -5, 0, 5, 10, \ldots\}$

$1 + H = \{\ldots -9, -4, 1, 6, 11, \ldots\}$

$2 + H = \{\ldots -8, -3, 2, 7, 12, \ldots\}$

$3 + H = \{\ldots -7, -2, 3, 8, 13, \ldots\}$

$4 + H = \{\ldots -6, -1, 4, 9, 14, \ldots\}$

Any other cosets $n + H$ coincides with one of the above cosets.

$\therefore Z/H = \{0+H, 1+H, 2+H, 3+H, 4+H\}$

(ii) Here, H and Z both are infinite sets

$\therefore$ Index of H in $Z = i_Z(H) =$ the number of distinct cosets $= 5$.

$\therefore$ the index of H in Z is 5.

Lagrange's Theorem

If G is a finite group and H a subgroup, then the order of H is a divisor of the order of G.

Note:- The converse of Lagrange's theorem need not be true.
i.e. if m divides $o(G)$, then G need not have a subgroup of the order m.

→ If G is finite group and $a \in G$, then $o(a)$ divides $o(G)$.

→ If G is finite group and $a \in G$ then $a^{o(G)} = e$.

Ex Let G be a group of the order p, where p is a prime. Find all subgroups of G.

Sol By Lagrange's theorem, the order of subgroup of ~~G~~ H of G divides order of G. Hence, $|H| = 1$ or p. Thus, $\{e\}$ and G, itself are the only subgroup of G.